the iceberg. Here is what happened when a professor at St Joseph's, a small Indiana college, decided to work towards the goal of 'Android for everyone'! Using low-cost, refurbished or used Android phones, a small team of computer science students at the college started an outreach program for the local community. The team used the Android smartphones as the primary target platform for introductory programming classes, started programmes to collect historical images with embedded geo-location and sensor-provided data, offered real-time field access to USDA soil surveys and classification data, and enabled the recording of audio and video for a variety of media-distribution purposes. The team loaned the phones to economically-challenged people through a local café, whose owner provided free Wi-Fi to clients. So, the phones served multiple purposes for the local population—as sensors, media and data collection tools, and as a gateway to the WWW and its treasures.

The success of Beaglebone is another notable example of the versatility of Android. BeagleBone is an embedded computer platform based on an ARM Cortex-A8 processor that runs Android 4.0 and Ubuntu software. It has been used for a range of interesting projects such as underwater robots, 3D printers, a dirty dish detector and a real-life Iron Man suit! The Android platform is also popular for in-car infotainment systems. Parrot Asteroid was an early example. More recently, Clarion launched AX1 and Mirage, which run Android 2.3.7 and 2.2 respectively and feature GPS-based navigation, a 16.5 cm (6.5-inch) screen and options for wireless data access.

So we could say that the most significant recent milestone in Android's journey is its acceptance in the enterprise application space. A survey conducted by cross-platform tool company Appcelerator in Q3 of 2013 indicated that the enterprise arena is slipping away from Microsoft, while acceptance for Android is growing and iOS is the number one priority for IT decision makers in enterprises. In a related press report, Nolan Wright, co-founder and CTO at Appcelerator, mentioned that one of the reasons for the increasing interest in Android could be Android's strong overall market share, and with the current bring-your-own-device (BYOD) culture, enterprises have to build apps for multiple platforms.

Sharing the BIG stuff

Trend: If cloud computing is one way of sharing, high-performance computing (HPC) resource sharing is another. A lot of HPC infrastructure is Linux-based, and predominantly open in nature, which enables it to be shared by more than one research team. While some are owned by the government and shared by the sub-agencies, some are owned by universities and shared by the labs, and yet others are rented out by service providers. In any case, it would be a sin to spend so much on a HPC setup and not share it. So, 'open' is the way to go!

Trend+: Universities such as Yale have several HPC clusters available for faculty research. These clusters are a shared Linux environment with most of the popular applications, compilers and programs to support intensive research. So do research organisations like the European Organisation for Nuclear Research (CERN). Last year's hyped news about CERN researchers discovering a sub-atomic particle travelling faster than light was also the result of resource sharing to crunch Big Data at CERN. Its HPC setup enabled thousands of scientists from over 100 countries to come together to collaborate on path-breaking research—something that would have been impossible without an open platform. They used Apache Hadoop running on an open cloud and grid infrastructure, to analyse over 20 million GB of data produced each year by a mammoth structure called the CERN Large Hadron Collider, the world's largest and most powerful particle accelerator. Such open platforms can prove beneficial for other fields of research too, especially drug discovery and medical research, which is often carried out in secret, by industry leaders!

Wear your tech

Trend: From being carried in bags, now technology has evolved to a state where it is being worn. And we are not just talking about medical aids like pacemakers, but about trendy watches, pendants, wristbands and other such accessories, which have embedded systems performing significant tasks. Wearable devices help in a variety of ways—to monitor a person's health; evaluate their exercise routine, diet or sleep patterns; track their location, deliver new gaming or travel experiences through augmented reality, for communication, or just to run simple applications.

Trend+: There is a lot of wearable technology being used by the army. In fact, soldiers' jackets are now smarter than many of our phones! The other key application areas are fitness and healthcare. Some of the popular products in this category are Nike Fuelband, Fitbit and MyBasis. Slowly, wearable devices are being used in industries as well. For example, Google glass-based devices for miners and architects, or touch-sensitive gloves for surgeons. Other wearable devices help caretakers keep an eye on children or the elderly, by tracking their movements.

Generic or otherwise, wearable devices are a challenge to design. Designers have to always remember that the device has to be worn by a human, comfortably. It should neither be too bulky nor too hot. Similarly, it should be safe. These basic requirements greatly influence the design goals of wearable tech devices.

Wearable technology requires very low-energy, battery-powered components to minimise heat dissipation and provide extended battery life. Size is another factor that needs to be taken into consideration. For instance, the